

AI-based Urine Microscopy Image Analysis for Predicting Urinary Tract and Renal Diseases

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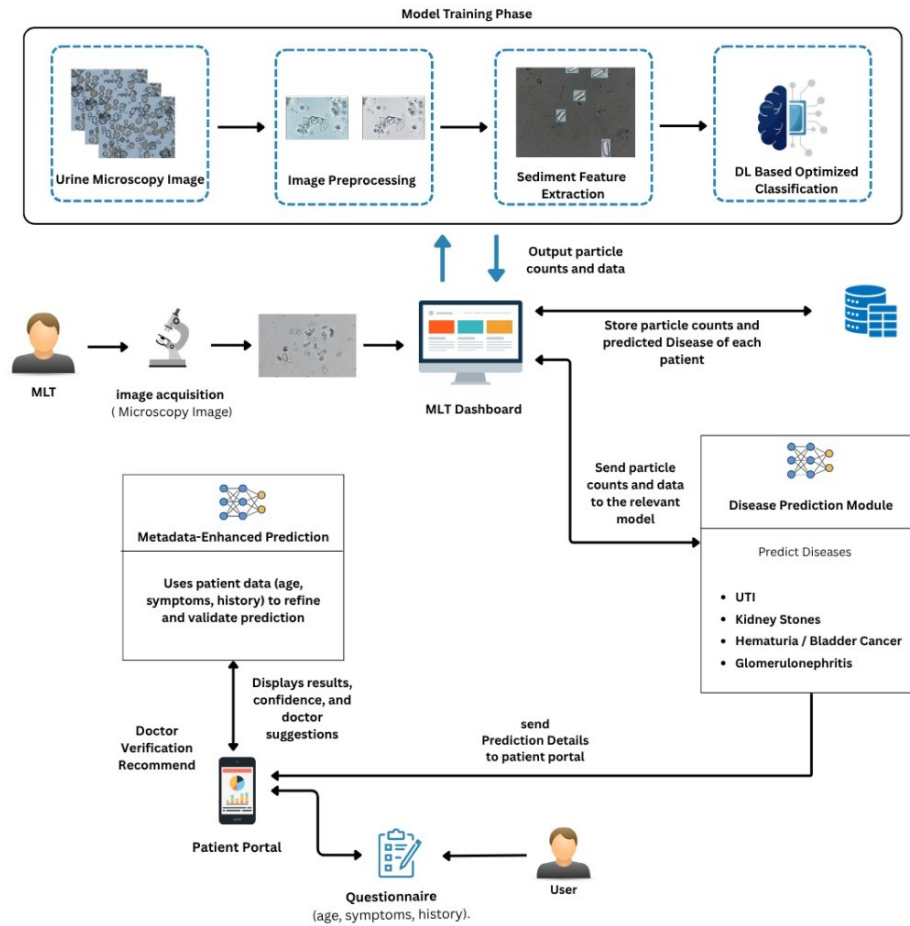
1. Introduction

Urinalysis is an important diagnostic procedure used in clinical laboratories to identify urinary tract infections (UTIs), kidney diseases, and other renal abnormalities. Microscopic examination of urine sediment enables the observation of particles such as white blood cells (WBCs), red blood cells (RBCs), bacteria, yeast cells, casts, and crystals. The presence and morphology of these components provide important diagnostic information related to infections, hematuria, kidney stone formation, and renal pathologies. However, manual microscopic examination is a time-consuming and operator-dependent process that can lead to inconsistencies in diagnosis and increased workload for laboratory professionals.

Recent advancements in artificial intelligence and deep learning have enabled the automation of complex image analysis tasks in medical imaging. Convolutional neural networks and object detection architectures have demonstrated high performance in detecting and classifying microscopic particles in biomedical images. Despite these advancements, most existing automated urinalysis systems focus on isolated tasks such as detecting individual particles or predicting diseases through black-box models without integrating clinical reasoning.

To address these limitations, this research proposes an AI-based microscopic image analysis platform that integrates multiple deep learning models to detect and analyze various urine sediment components. The system aims to automate the identification of microscopic particles and combine this information with clinically informed rules to generate diagnostic insights related to urinary tract infections, kidney stone risk, hematuria causes, and renal diseases. By combining multiple specialized models into a unified framework, the platform improves diagnostic reliability, reduces manual workload, and enables faster clinical decision support.

2. System Architecture



The proposed system follows a modular architecture consisting of multiple deep learning components integrated into a unified analysis pipeline. The architecture is designed to process microscopic urine images, detect and analyze various sediment particles, and generate clinically relevant diagnostic outcomes.

The workflow of the system begins when a user uploads a microscopic urine image through the web interface. The image is transmitted to the backend server, where preprocessing operations such as resizing, normalization, and augmentation are applied to prepare the image for model inference.

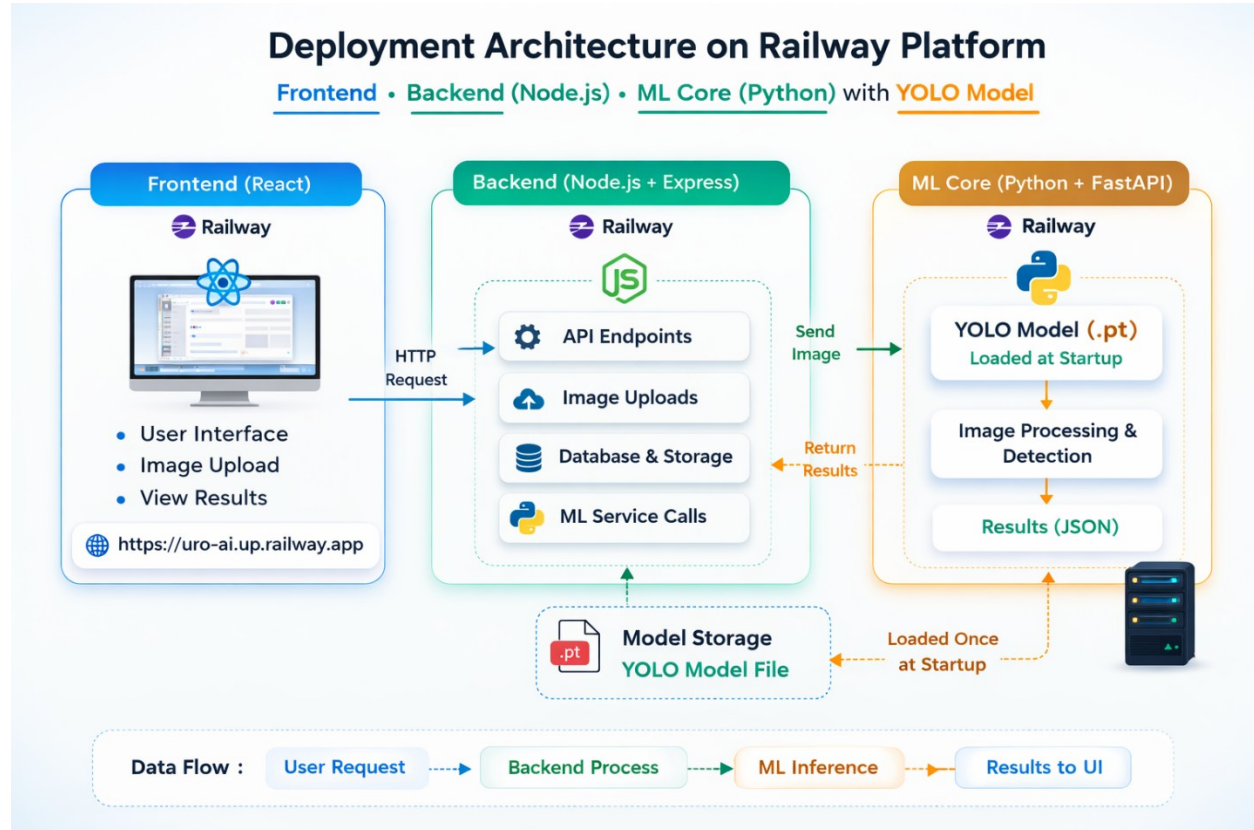
The processed image is then passed through multiple specialized AI modules responsible for analyzing different urine sediment components. These modules include particle detection models, segmentation models, and classification models. Each module independently identifies and analyzes specific microscopic elements such as WBCs, RBCs, yeast cells, bacteria, crystals, and urinary casts.

The outputs generated by these models include detected particle locations, segmentation masks, classification labels, and particle counts. These results are aggregated and passed to a rule-based decision layer that interprets the findings using clinically informed diagnostic logic. Based on the detected microscopic evidence, the system predicts potential conditions such as bacterial UTI, fungal UTI, kidney stone risk, hematuria origin, and renal disease indicators.

Finally, the processed results are returned to the frontend application where users can visualize detected particles, segmentation overlays, and predicted diagnostic outcomes.

3. Deployment Environment

The AI-based urine microscopy analysis system was deployed using a cloud-based infrastructure to ensure accessibility, scalability, and reliability.



The frontend application was developed using **React.js**, which provides a responsive and interactive user interface for uploading microscopy images and viewing AI analysis results.

The backend server was developed using **Node.js and Express.js**, which handles API communication, image uploads, system logic, and interaction with the AI models.

The machine learning inference service was implemented using **Python and FastAPI**, which executes the trained deep learning models for microscopic particle detection and classification.

The system was deployed using the **Railway cloud platform**, which allows automatic deployment from a GitHub repository and provides monitoring, logging, and service management features.

To ensure uploaded microscopy images are preserved even during server restarts, a **persistent storage volume** was configured in Railway.

4. Deployment Tools and Platforms

Several modern tools and technologies were used for system development and deployment.

Railway Cloud Platform

Railway was used as the main cloud deployment platform. It allows applications to be deployed directly from a GitHub repository and supports automatic builds, containerized services, environment variable management, and persistent storage.

GitHub

GitHub was used as the version control system for managing the project source code. The repository contains both frontend and backend codebases. Railway automatically deploys updates whenever changes are pushed to GitHub.

React.js

React.js was used to build the frontend interface. It enables users to upload urine microscopy images and view the AI analysis results through a web-based interface.

Node.js and Express

The backend server was implemented using Node.js and the Express framework. This server processes API requests, manages image uploads, and coordinates communication with the machine learning service.

Python FastAPI

FastAPI was used to implement the machine learning inference service responsible for executing deep learning models.

Persistent Storage

Railway persistent storage volumes were configured to store uploaded microscopy images.

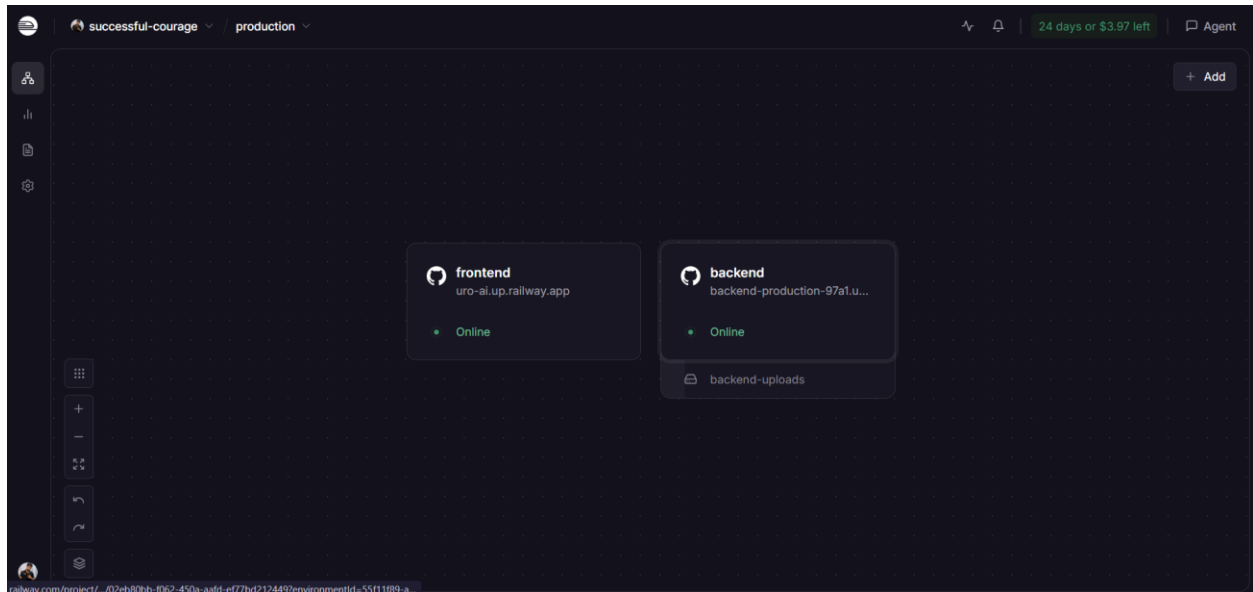


Figure 2: Railway Deployment Dashboard Showing Running Frontend and Backend Services

Deployment Process

The deployment process followed several steps to ensure proper integration of all system components.

First, the frontend and backend applications were prepared for deployment. The React frontend was configured to communicate with the backend API through environment variables.

Next, the complete project source code was pushed to GitHub. The Railway platform was then connected to the GitHub repository to enable automatic builds and deployments.

Three separate services were deployed on Railway:

- Frontend Service – hosts the React application
- Backend Service – runs the Node.js server
- ML Core Service – runs the Python FastAPI AI model

The backend service was configured with a persistent storage volume named **backend-uploads** to store uploaded microscopy images.

Environment variables were then configured to enable communication between services.

After configuration, Railway automatically built and deployed the services. Deployment logs were monitored to verify that all services started successfully.

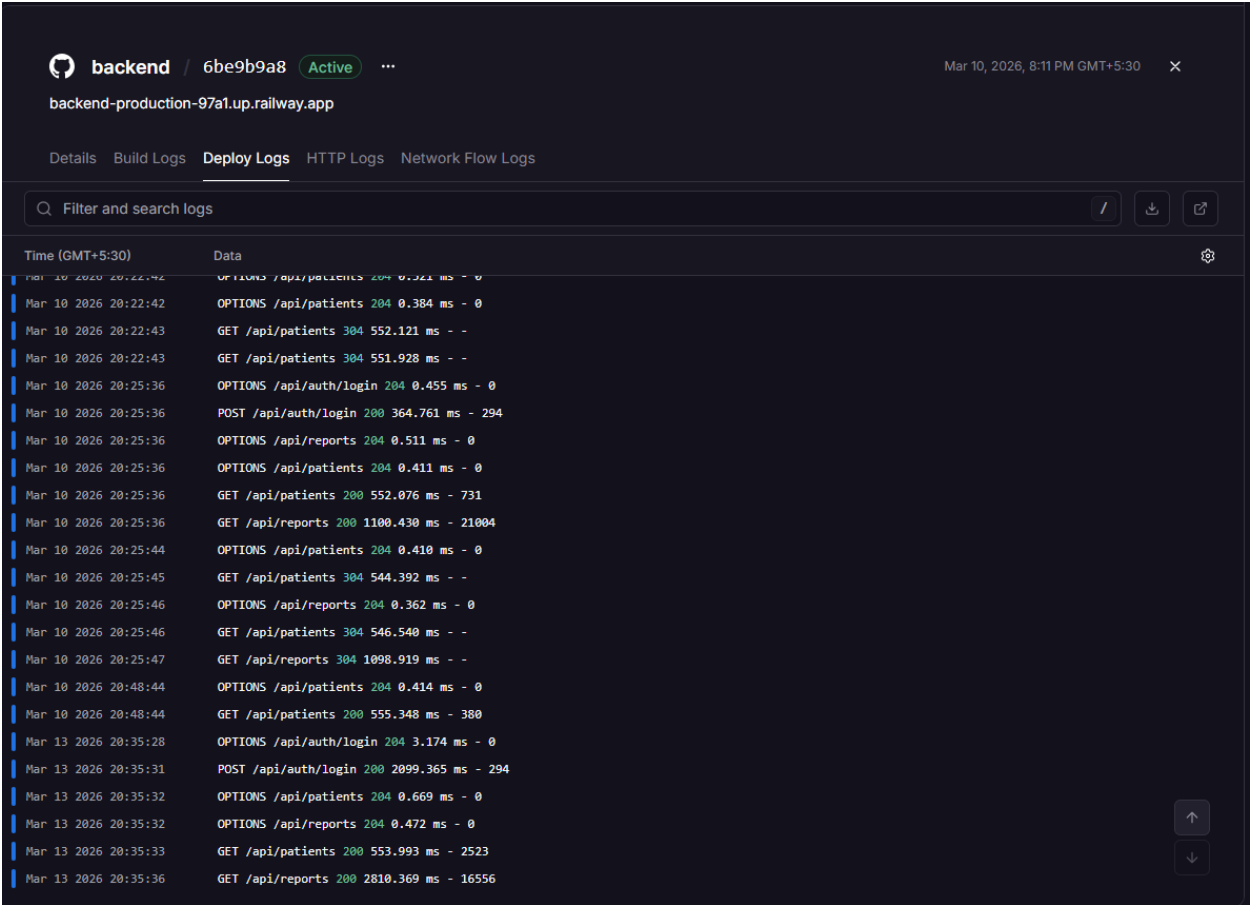


Figure 3: Backend Deployment Logs and Service Configuration on Railway

5. Model Integration

The proposed platform integrates multiple deep learning models designed to perform specialized analysis tasks on urine microscopy images. Each module focuses on detecting or classifying specific microscopic particles relevant to clinical diagnosis.

For urinary tract infection analysis, white blood cells are detected using a YOLO-based object detection model, which provides efficient small-object detection and real-time inference. To obtain detailed morphological information, a segmentation model based on UNet++ with an EfficientNet encoder is used to generate pixel-level masks of WBC regions. Yeast cells are also detected using a YOLO-based model, while segmentation is performed using a transformer-based architecture to refine the boundaries of densely clustered yeast cells.

Due to the lack of precise bounding box annotations for bacteria in microscopy images, bacterial analysis is implemented as a classification task. A convolutional neural network based

on EfficientNet is trained to classify cropped microscopic images into bacterial or non-bacterial categories. The outputs from these models are combined to support automated UTI classification.

In addition to infection detection, the system includes modules for crystal analysis, red blood cell morphology analysis, and urinary cast detection. Crystal analysis is performed using a two-stage approach where object detection is first used to localize crystal regions, followed by a convolutional neural network classifier that identifies crystal subtypes related to kidney stone risk.

Red blood cell analysis involves detecting individual RBCs from microscopic images and classifying them into isomorphic or dysmorphic types using an EfficientNet-based model. This classification helps determine whether hematuria originates from glomerular or non-glomerular sources.

Urinary cast detection is implemented using an object detection model to identify casts within the microscopic images. Cropped cast regions are then classified into clinically significant subtypes using convolutional neural networks. The outputs from all modules are integrated through a rule-based diagnostic layer that interprets particle counts and classifications to generate disease predictions.

6. System Access and Usage

The microscopic image analysis platform is designed as a web-based application that allows users to analyze urine microscopy images through an intuitive interface. The system consists of a frontend web application and a backend server responsible for executing the AI models.

Users access the platform through a web browser and upload microscopic urine images obtained from laboratory microscopy equipment. Once an image is uploaded, it is transmitted to the backend server through an API request. The backend system processes the image and performs model inference using the integrated deep learning modules.

The AI models analyze the image to detect microscopic particles, segment important structures, and classify relevant biological entities. The detected results are then processed through a diagnostic interpretation layer that evaluates the particle counts and classifications to determine potential disease indicators.

After processing is completed, the system returns the analysis results to the user interface. The results include visual overlays showing detected particles, segmentation masks highlighting cellular structures, and summarized diagnostic insights related to urinary tract infections, kidney stone risk, hematuria causes, and renal disease indicators.

This interactive platform enables laboratory professionals, researchers, and clinicians to rapidly analyze urine microscopy images and obtain automated diagnostic insights, improving efficiency and reducing reliance on manual microscopic examination.

7. Future Improvements

Although the proposed system demonstrates strong performance in automated urine microscopy analysis, several improvements can further enhance its accuracy, scalability, and clinical applicability.

One key improvement involves expanding the dataset used to train the deep learning models. Increasing the diversity and volume of annotated microscopy images will improve model generalization and enable better performance in real-world laboratory environments.

Another potential improvement is the integration of more advanced model architectures and ensemble techniques to improve detection accuracy for challenging particles such as rare urinary cast subtypes and dysmorphic red blood cells. Enhancing segmentation precision and improving class balance within datasets will also contribute to more reliable predictions.

Future work may also involve deploying the platform in real clinical laboratory environments to evaluate its performance in real-time diagnostic workflows. Integration with laboratory information systems and digital microscopy devices could further streamline automated urinalysis.

Additionally, incorporating cloud-based infrastructure and scalable deployment frameworks will enable the system to handle larger volumes of microscopic image data and support multi-institutional clinical use. These improvements will help transform the proposed framework into a robust clinical decision-support system for automated urine microscopy analysis.